1.

Write a program in C to simulate 20 rolls of a six-sided dice and print the value of each roll.

Sample Output				
6	6	5	5	6
5	1	1	5	3
6	6	2	4	2
6	2	3	4	1

2.

The university gives passwords to users. Every student is given an 8 characters long random password. The passwords have the following format:

The first 3 chars are random uppercase English letters

Following 3 chars are random lowercase English letters

The last 2 chars are random digits (from 0-9)

Question:

Write a program that produces n random passwords. **Solve it using at least 3 functions.**

Input specification

You will be given just one integer (n) where n is between 1 and 1000.

Output specification

Show n passwords with the given format above in n lines. All passwords must be different than all others.

Note:

1. Use rand() function.

2. Lowercase alphabet characters in ASCII have values ranging from 97 to 122, while uppercase alphabet characters have values ranging from 65 to 90.

Sample Input	Sample Output
Enter the number of passwords to generate: 5	DBEshz53 CSMwqa45 FSJmlx39 LJTxzw82 XUWhtl85

3.

Write a program in C that prompts the user to enter an integer then displays its reversal. **Solve it using 2 functions.**

Sample Input	Sample Output
3456	6543

4.

Write a program in C that prompts the user to enter a hexadecimal number as a string and convert it into a decimal. **Solve using 1 function.**

Sample Input	Sample Output
Enter a hex number: **AB8C**	The decimal value for hex number AB8C is 43916

Enter a hex number: **af71**	The decimal value for hex number af71 is 44913

5.

Write a program in C that prints the first 50 prime numbers. **Solve it using 1 function, for determining whether a number is a prime. Extra: use another function for printing the prime numbers.**

Sample Output
The first 50 prime numbers are:

2	3	5	7	11	13	17	19	23	29
31	37	41	43	47	53	59	61	67	71
73	79	83	89	97	101	103	107	109	113
127	131	137	139	149	151	157	163	167	173
179	181	191	193	197	199	211	223	227	229

6.

You will be first given one number (n) which is followed by two series of integers (n space separated values). Calculate the sum of two series and show them.

Input specification

You will be given one integers (n) in the first line, and $1 \leq n \leq 10.000$. The following two lines contain a two series of integer numbers a[n] and b[n] where $-1000 \leq a[n], b[n] \leq 1000$.

Output specification

You will give one serie, which is the sum serie of two input series.

Sample Input	Sample Output
5 3 6 9 5 4 2 5 6 2 7	5 11 15 7 11

7.

You will be first given one number (n) which is followed by one series of integers (n space separated values). Calculate the difference between max and min in the array.

Input specification

You will be given one integer (n) in the first line, and $1 \leq n \leq 10.000$. The following line contains a series of integer numbers a[n] where $-10000 \leq a[n] \leq 10000$.

Output specification

Show the difference between max and min element in the series.

Sample Input	Sample Output
5 3 6 9 5 4	6
6 5 8 4 -9 3 2	17

8.

In research, your professor has realized that the data he is working on has some

noise. For this reason, he wants to subtract the smallest value of every row from the row elements. As a result, the smallest number of every row will be 0.

Question: Write a program that is going to get n rows of m numbers. Then, the program will subtract the smallest value of every row from the row elements.

Input specification

In the beginning, you will be given 2 integers:

• the number of rows (n): an integer between 0 and 5000

• the number of numbers in every row (m): an integer between 0 and 1000 then in the following n lines you will be given m integers which are between -10000 and 10000.

Output specification

Show m integers in n lines.

Sample Input	Sample Output
3 6 5 6 1 6 1 7 10 5 15 24 13 12 9 4 16 6 7 15	4 5 0 5 0 6 5 0 10 19 8 7 5 0 12 2 3 11

9.

(Sales Commissions) Use a one-dimensional array to solve the following problem. A company pays its salespeople on a commission basis. The salespeople receive $200 per week plus 9% of their gross sales for that week. For example, a salesperson who grosses $3,000 in sales in a week receives $200 plus 9% of $3,000, or a total of $470. Write a C program (using an array of counters)

that determines how many of the salespeople earned salaries in each of the following ranges (assume that each salesperson's salary is truncated to an integer amount):

a) $200–299

b) $300–399

c) $400–499

d) $500–599

e) $600–699

f) $700–799

g) $800–899

h) $900–999

i) $1000 and over

Sample Input	Sample Output
Enter the number of salespeople: 5 $3000 $5000 $1500 $2500 $7500	$200 - $299 0 $300 - $399 1 $400 - $499 2 $500 - $599 0 $600 - $699 1 $700 - $799 0 $800 - $899 0 $900 - $999 0 $1000 and over 0

10.

Use one-dimensional arrays to solve the following problem. Read in two sets of numbers, each having 10 numbers, each of them in the range from 1-10. After reading all the values, display all the unique elements in the collection of both sets of numbers. Use the smallest possible array to solve this problem.

Sample Input	Sample Output
Enter 10 numbers for the first set: 1 1 1 1 2 3 4 5 6 7 Enter 10 numbers for the second set: 5 5 5 6 7 8 9 10 10 10	Unique numbers in both sets: 2 3 4 8 9